

CSX3007 Home Assignment1

1. How many different numbers can be represented with 16 bits?
2. What is the largest unsigned 32-bit binary number?
3. What is the largest 16-bit binary number that can be represented with
 - 3.1 Unsigned numbers?
 - 3.2 Two's complement numbers?
 - 3.3 Sign/magnitude numbers?
4. Convert the following hexadecimal numbers to decimal. Show your work.
 - 4.1 $A5_{16}$
 - 4.2 $3B_{16}$
 - 4.3 $FFFF_{16}$
 - 4.4 $ED3A_{16}$
5. Repeat Exercise 4, but convert to unsigned binary.
6. Convert the following two's complement binary numbers to decimal.
 - 6.1 1010_2
 - 6.2 110110_2
 - 6.3 01110000_2
 - 6.4 10011111_2
7. Repeat Exercise 6, but convert to 8-bit sign/magnitude numbers.
8. Convert the following 4-bit two's complement numbers to 8-bit two's complement numbers.
 - 8.1 0101_2
 - 8.2 1010_2
9. Perform the following additions of unsigned binary numbers. Indicate whether or not the sum overflows an 8-bit result.
 - 9.1 $10011001_2 + 01000100_2$
 - 9.2 $11010010_2 + 10110110_2$
10. Repeat Exercise 9, assuming that the binary numbers are in two's complement form.
11. Convert the following decimal numbers to 8-bit binary numbers and add them. Indicate whether or not the sum overflows a 8-bit result.
 - 11.1 $27_{10} + 31_{10}$
 - 11.2 $-4_{10} + 19_{10}$
 - 11.3 $3_{10} + -32_{10}$
 - 11.4 $-16_{10} + -9_{10}$

12. Perform the following additions of unsigned hexadecimal numbers.
Indicate whether or not the sum overflows an 8-bit (two hex digit) result.
- 12.1 $7_{16} + 9_{16}$
 - 12.2 $13_{16} + 28_{16}$
 - 12.3 $AB_{16} + 3E_{16}$
 - 12.4 $8F_{16} + AD_{16}$
13. In a binary-coded decimal (BCD) system, four bits represent a decimal digit from 0 to 9. For example, 3710 is written as 00110111BCD.
- 13.1 Write 289_{10} in BCD.
 - 13.2 Convert 100101010001BCD to decimal
 - 13.3 Convert 01101001BCD to binary
 - 13.4 Show BCD adjust operations on $92_{10} + 25_{10}$
 - 13.5 Show BCD adjust operations on $38_{10} + 22_{10}$
 - 13.6 Show BCD adjust operations on $78_{10} + 32_{10}$
14. How many different truth tables exist for Boolean functions of N -variables?
15. Create the following logic functions out of NAND gates:
- 15.1. NOT
 - 15.2. AND
 - 15.3. OR
 - 15.4. NOR
 - 15.5 XOR
 - 15.6 XNOR
16. Show a 3-input XOR gate with the help of its Truth Table.
17. Show how a 2-input XOR gate can act as a NOT and BUFFER function.